2. Interchanging bras and kets corresponds to complex conjugation:

$$\langle B|A\rangle = \langle A|B\rangle^*.$$

Exercise 1.1:

a) Using the axioms for inner products, prove

$$\{\langle \mathbf{A}| + \langle \mathbf{B}|\}\,|\mathbf{C}\rangle = \langle \mathbf{A}|\mathbf{C}\rangle + \langle \mathbf{B}|\mathbf{C}\rangle.$$

b) Prove <**A**|**A**> is a real number.

In the concrete representation of bras and kets by row and column vectors, the inner product is defined in terms of components:

$$\langle B|A\rangle = \begin{pmatrix} \beta_1^* & \beta_2^* & \beta_3^* & \beta_4^* & \beta_5^* \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{pmatrix}$$

$$= \beta_1^*\alpha_1 + \beta_2^*\alpha_2 + \beta_3^*\alpha_3 + \beta_4^*\alpha_4 + \beta_5^*\alpha_5. \quad (1.2)$$

The rule for inner products is essentially the same as for dot products: add the products of corresponding components of the vectors whose inner product is being calculated.